

Programme	B. Sc. Computer Science				
Course Title	Fundamentals Of Programming (C Language)				
Type Of Course	Major				
Semester	II				
Academic Level	100 – 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamentals of Algorithms and Flowcharts 2. CSC1CJ101 – Fundamentals of Computers and Computational Thinking				
Course Summary	The objectives of this course are to make the student understand programming language, programming, concepts of Loops, reading a set of Data, stepwise refinement, Functions, Control structure, Arrays, Structures, Unions, and Pointers. After completion of this course the student is expected to analyze the real life problem and write a program in ‘C’ language to solve the problem. The main emphasis of the course will be on problem solving aspect i.e. developing proper algorithms.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Remember the program structure of C with its syntax and semantics	U	C	Instructor-created exams / Quiz
CO2	Use the various constructs of a programming language viz. conditional, iteration and recursion.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Implement the algorithms in C language.	Ap	P	Practical Assignment / Observation of Practical Skills
CO4	Use simple data structure like array in solving problems.	Ap	C	Practical Assignment / Observation of Practical Skills
CO5	Handling pointers and memory management functions in C.	Ap	P	Practical Assignment / Observation of Practical Skills
CO6	Develop efficient programs for solving a problem.	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs
I	Introduction to C Language (12 Marks)		10
	1	History of C, Importance of C, and sample programs	2
	2	Character set, Tokens, Constants, Variables, and Data types	2
	3	Operators - Arithmetic, Relational, logical, assignment, increment, decrement, conditional, bitwise and special operators. Arithmetic expressions, operator precedence, type conversions, mathematical functions	3
	4	Managing Input and Output Operators: Reading and writing a character, formatted input, formatted output.	3
II	Decision Making Branching and Looping (18Marks)		10
	5	Decision making with If - simple If, If else, nested If else, else If ladder	3
	6	Switch statement, conditional operator, Goto statement	2
	7	Loops: while, do while, for statements and nested loops	3
	8	Jumps in loops – break, continue	2
III	Arrays and Functions (20 marks)		15
	9	One dimensional array – declaration, initialization and accessing	2
	10	Two dimensional array – declaration, initialization and accessing	2
	11	Multi dimesnsional array, dynamic array	1
	12	Strings – Reading, Writing. Arithmetic operations on characters, Comparisons and string handling functions	2
	13	Functions – Need, Elements of user defined functions and definition	2
	14	Return values and their types, function call and declaration, call by value and call by reference	2
	15	Categories of functions, Nesting of functions	1
	16	Recursion and command line arguments	1
	17	Passing arrays to functions and passing strings to functions	2
IV	Storage Classes, Structure and Union, Pointers (20 Marks)		10
	18	Storage classes – The scope, visibility and lifetime of variables. Auto, Extern, Static and Register storage classes. Storage classes in a single source file and multiple source files	2
	19	Structure and Union - Defining, giving values to members, initialization and comparison of structure variables, arrays of structure, arrays within structures, structures within structures, structures and functions, unions	2
	20	Pointers definition, declaring and initializing pointers, accessing a variable through address and through pointer, pointer expressions, pointer increments and scale factor	2
	21	Pointers and arrays, pointers and functions, pointers and structure	2
	22	Dynamic memory allocation and memory management functions	2
V	Hands-on Problem Solving Using C Practical Applications, Case Study and Course Project		30

1	Implement the following: 1. Variables, Data types, Constants and Operators: 1.Evaluation of expression ex: $((x+y)^2 * (x+z))/w$ 2.Temperature conversion problem (Fahrenheit to Celsius) 3.Program to convert days to months and days (Ex: 364 days = 12 months and 4 days) 4. Salesman salary (Given: Basic Salary, Bonus for every item sold, commission on the total monthly sales) 2. Decision making (Branch / Loop) Statements: 5.Solution of quadratic equation 6.Maximum of three numbers 7.Calculate Square root of five numbers (using goto statement) 8.Pay-Bill Calculation for different levels of employee (Switch statement) 9. Fibonacci series 10.Armstrong numbers 11.Pascal 's Triangle 3. Arrays, Functions and Strings: 12.Prime numbers in an array 13.Sorting data (Ascending and Descending) 14.Matrix Addition and Subtraction 15.Matrix Multiplication 16.Transpose of a matrix 17Function with no arguments and no return value 18. Functions with argument and return value 19.Functions with argument and multiple return values 20.Function that convert lower case letters to upper case 21. Factorial using recursion. 22. Perform String Operations using Switch Case	30
	23. Largest among a set of numbers using command line argument 4. Structures and Union: 24. Structure that describes a Hotel (name, address, grade, avg room rent, number of rooms) Perform some operations (list of hotels of a given grade etc.) 25. Using Pointers in Structures. 26. Cricket team details using Union. 5. Pointers: 27.Evaluation of Pointer expressions 28.Function to exchange two pointer values 29. Reverse a string using pointers 30.Insertion, deletion, and searching in an array	

Mapping of COs with PSOs and POs:

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	-	1	1	-	-						
CO 2	-	1	2	2	-	-						
CO 3	-	1	3	3	-	-						
CO 4	1	1	2	2	-	-						
CO 5	-	2	2	2	-	-						

CO 6	-	1	3	3	1	1						
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6			✓	